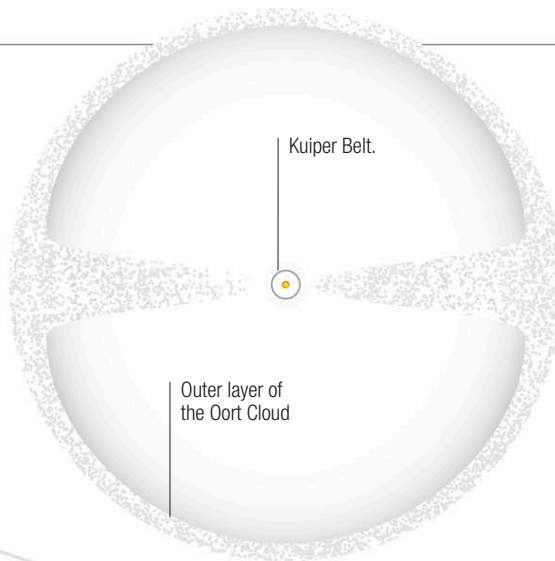
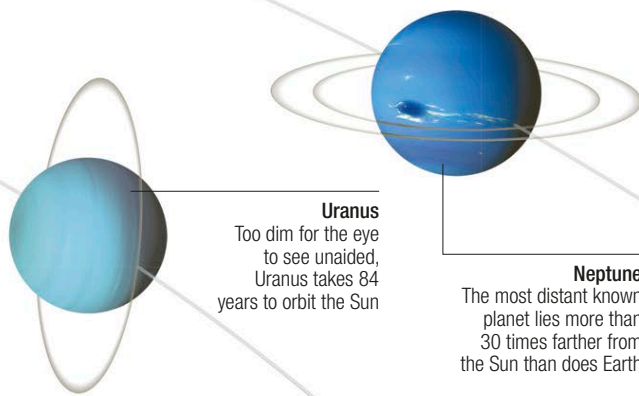




◀ The Oort Cloud

Billions of the minor planets were thrown out of the Solar System by the gravity of the newly formed planets. These minor planets now form this vast cloud of comets that may stretch one-quarter of the way to the nearest star.



Uranus

Too dim for the eye to see unaided, Uranus takes 84 years to orbit the Sun

Neptune

The most distant known planet lies more than 30 times farther from the Sun than does Earth

The term **planet** originates from the ancient Greek term for **wandering star**

Rocky planets

The planets nearest the Sun—Mercury, Venus, Earth, and Mars—contain far more rock and metal than gas. Each one has a surface shaped by volcanoes, where hot, molten material from the interior has broken through the solid crust. Apart from Mercury, these planets possess a significant atmosphere that has protected its surface, to varying degrees, from impacts. These atmospheres may have largely originated in asteroid and comet impacts, with the latter possibly also delivering large quantities of water to the planets. Only Earth and Mars have moons. Mars's natural satellites, Phobos and Deimos, are believed to be captured asteroids.

Gas planets

The four outer worlds—Jupiter, Saturn, Uranus, and Neptune—are bloated giants largely formed of gas surrounding a dense core, each accompanied by a large retinue of moons. When young, these planets grew large enough to draw in gas from the surrounding nebula. The motions of their churning atmospheres are driven by internal heat as well as energy from sunlight. Given their cold environments far from the Sun, these planets' many moons are predominantly worlds with water ice crusts, some heated inside due to the effects of tides. Some of these satellites have atmospheres, and others have active volcanoes.

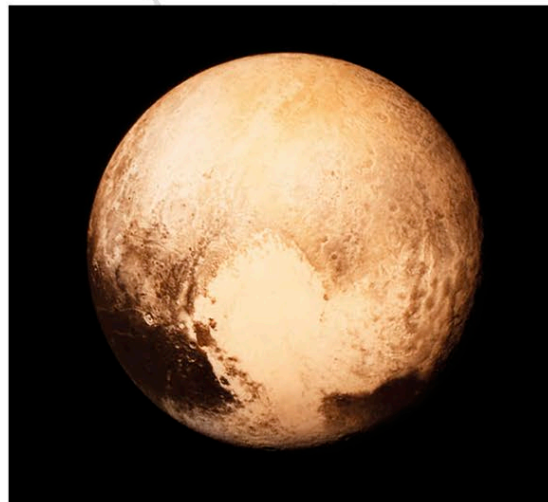
Minor bodies

In the early Solar System, dust and ice grains first formed small bodies, and these then accumulated to form the planets. Billions of the small bodies did not, however, become parts of planets, and they remain as minor bodies today. These are the asteroids and comets, and studying their make-up can tell us much about conditions in the early Solar System. The smallest of these—some are mere dust grains—enter our atmosphere, appearing as meteors. Many larger minor bodies, however, present Earth with a threat should a collision occur. Their rare impacts can potentially lead to global disruption, as is thought to have occurred 65 million years ago, leading to the extinction of the dinosaurs.



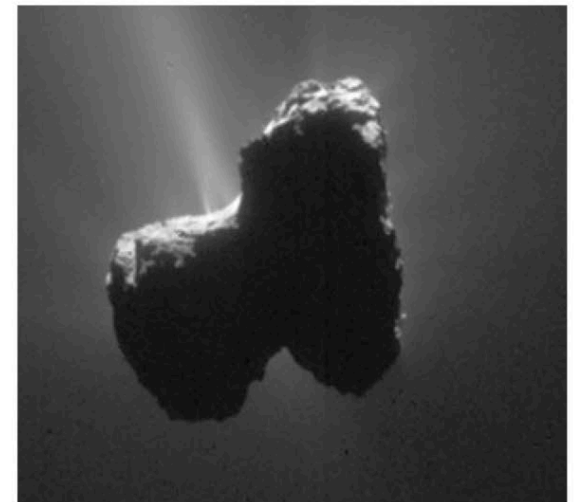
△ Asteroids

Most of these bodies, such as 951 Gaspra (shown here), are too small to form spheres. Despite being mostly rocky, several are now known to possess water ice under their surface.



△ Dwarf planets and Trans-Neptunian objects

Numerous icy bodies orbit beyond Neptune, in a flat disk known as the Kuiper Belt. Several of these are large enough to be classed as dwarf planets, including Pluto (shown here).



△ Comets

Comets, such as 67P/Churyumov-Gerasimenko (shown here), are small, icy bodies. As they approach the Sun, their ice turns into gas, releasing dust to create tails millions of miles long.